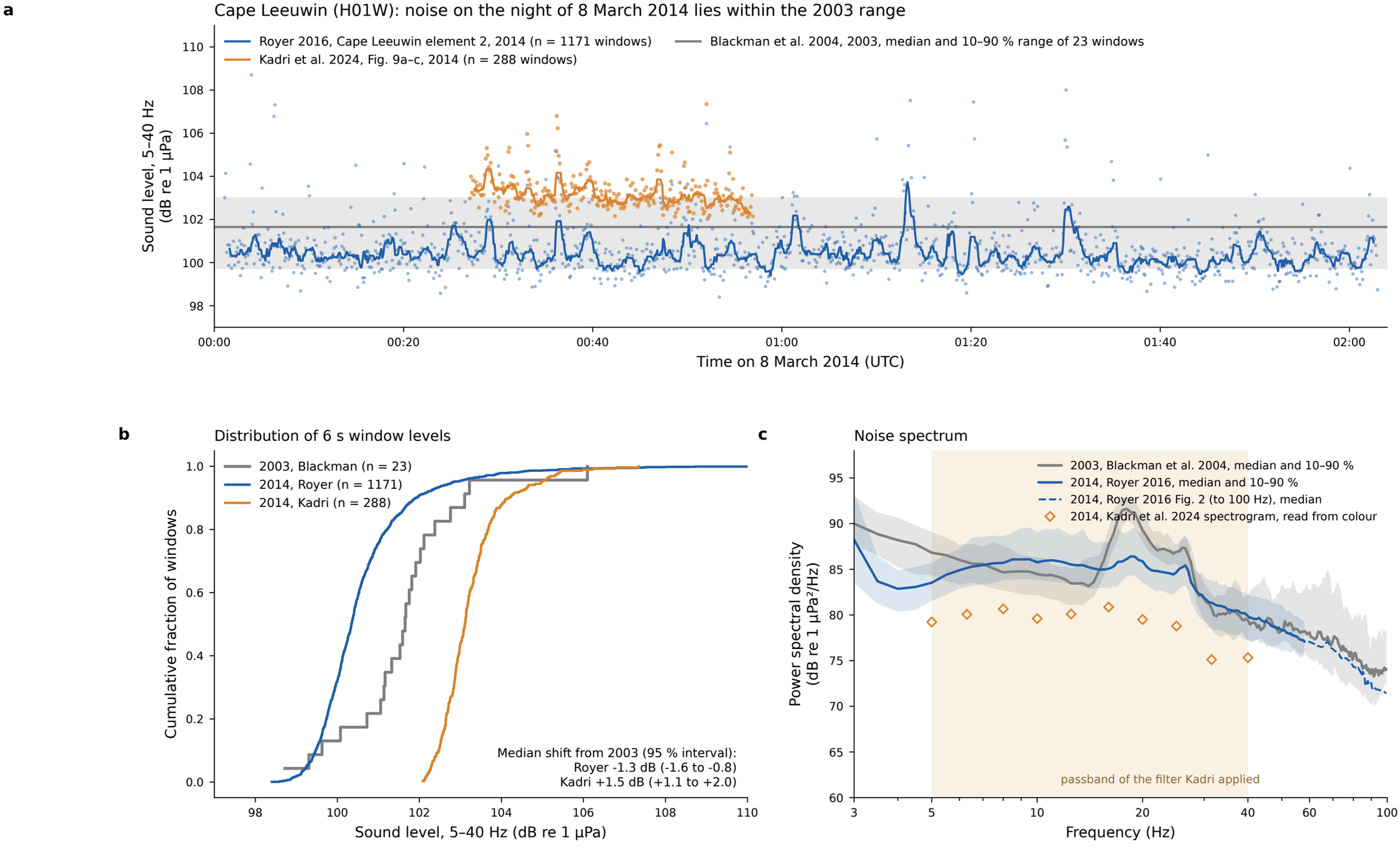


Plain note: The lines show the level of background sound at the Cape Leeuwin listening station. Grey is 2003, traced from Blackman et al. (2004). Blue and orange are the night of 8 March 2014, from two different published figures (Royer 2016; Kadri et al. 2024). The two 2014 sources differ by about 3 dB at the same time. The 2003 level lies between them. The 2003 spectrum is higher near 20 Hz. Whale calls are a possible cause but this has not been tested. These are digitised figures, not recordings.

Technical note: 5–40 Hz band level = integral of PSD over 5–40 Hz, per 6.25 s window. 2003: Blackman et al. 2004 (UCRL-TR-207323) App. B noise spectra (three-sensor mean, 6 s, 20 s before each 2003 shot), traced and calibrated with the FDSN IM.H01W1 response (prepare/blackman_noise.py; reproduced exactly); 23 quality-checked panels, May–June 2003. Royer: LDO-201608 appendix p. 18 spectrogram of H01W-2 (Hann 256, 80 % overlap, 250 Hz), colour inverted on a jet 70–100 dB scale (prepare/digitise_royer_2016.py), linear mean over 4 raster columns; 2 % of pixels at the 70 dB floor, 1.5 % at the 100 dB ceiling; 0–60 Hz; 60–100 Hz from Fig. 2 (00:15–01:45). Kadri: Fig. 9a–c vector traces (5 Hz high-pass plus 2–40 Hz band-pass, as stated by Kadri); vertices are min/max plot decimation (78 % local extrema), so RMS/1.954 (simulated Gaussian noise, 0.138 s bins, ±0.2 dB) gives σ; filter-equivalent level of the 2003 and Royer spectra differs from the plain 5–40 Hz level by < 0.15 dB (4th-order Butterworth assumed). Kadri spectrogram points: JPEG colour reading with a 64 dB colour floor and dropped dark pixels, biased low (earlier −5.96 dB offset). Shift = 2014 median − 2003 median; 95 % interval from a moving-block bootstrap (block = decorrelation lag, 3 windows) for 2014 and an iid bootstrap for 2003. Mann–Whitney, Kolmogorov–Smirnov and Anderson–Darling all give $p < 0.001$ on decorrelated samples (n = 391 and 96), but both shifts are within ±3 dB and inside the 2.8 dB disagreement between the two 2014 sources. Raster smoothing: the appendix strips read 1.1–1.4 dB below Royer's 3-minute panels at the same times, and their pixel standard deviation is 3.4 dB against 4.6 dB; a simulation of decibel-domain averaging of 3 spectrogram frames gives a −1.4 dB bias in the linear mean, so the Royer values shown are probably about 1.4 dB low (corrected median ≈ 101.7 dB). Values are shown uncorrected.